

Code-Layout, Readability and Reusability

Date	19 September 2026
Team ID	SWTID-2026-3833
Project Name	AI StudyBuddy API
Maximum Marks	2 Marks

Code Layout, Readability and Reusability

AI StudyBuddy follows a modular layout under `src/`, keeping every concern in its own folder so a new AI feature or route can be added without touching unrelated files.

Folder	Purpose	Reusability Practice
<code>utils/db.js</code>	MongoDB Atlas connection setup	Connection logic centralized once; the Express listener only starts after <code>connectDB()</code> resolves
<code>controllers/</code>	Request-handling logic per resource (auth, materials, admin)	Thin controllers delegate persistence to Models and all Gemini calls to <code>utils/gemini.js</code> , so AI-vendor logic isn't duplicated
<code>middleware/</code>	<code>auth.js</code> (JWT guard + RBAC), <code>upload.js</code> (multer file validation)	A single <code>protect / adminOnly</code> pair is reused across every protected and admin-only route instead of re-checking tokens/roles per controller
<code>models/</code>	Mongoose schemas (<code>User.js</code> , <code>Material.js</code>) with validation baked in	Required-field validation lives once in the schema, not repeated in every controller
<code>routes/</code>	Endpoint-to-controller mapping per resource	Each route file is mounted with its own prefix (<code>/api/auth</code> , <code>/api/materials</code> , <code>/api/admin</code>), keeping URL structure predictable
<code>utils/gemini.js</code>	A single <code>askGemini(prompt)</code> function wrapping the Google Gemini <code>gemini-2.5-flash</code> model	All four AI features (summary, flashcards, quiz, study plan) call the same function with a different prompt template, avoiding duplicated Gemini client setup

Consistent naming (*camelCase* functions, *PascalCase* models), *small single-purpose* controller functions, and comments above non-obvious logic (e.g., the Gemini prompt templates and the *multer* file filter) keep the codebase readable for all 5 contributors working in parallel.